

DATA 615 - Regression: Homework - 9

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1 Problem 9.18. Job Proficiency (p. 379. 30pts: a. 5, b. 15 (10 for the statistics, 5 for comments.), add part c. 10)

- In part a, ignore the statement “Use alpha limits of 0.05 and 0.10 for adding or deleting variable, respectively. Instead, use the default selection criterion in the software. Note that we will use”stepwise” in part (a).
- In part b, do the following:
 - Find/Compute R2, adjusted-R2, Mallow’s Cp, AIC, SBC (i.e., BIC), PRESS for the final model that you selected.
 - Find/Compute R2, adjusted-R2, Mallow’s Cp, AIC, SBC (i.e., BIC), PRESS for the regression model that includes the first-order terms of all 4 predictors.
 - Summarize your results.
- In addition to part (a) and (b), answer the following question.
 - c. Use R function regsubsets() to conduct the best subset selection. For those model comparison criteria can be found in the output, are their values the same as the ones you calculated in part (a)?

Refer to Job proficiency Problems 9.10 and 9.11:

A personnel officer in a governmental agency administered four newly developed aptitude tests to each of 25 applicants for entry-level clerical positions in the agency. For purpose of the study, all 25 applicants were accepted for positions irrespective of their test scores. After a probationary period, each applicant was rated for proficiency on the job. The scores on the four tests (X1, X2, X3, X4) and the job proficiency score (Y) for the 25 employees were as follows:

```
jp <- read.table("./data/CH09PR10.txt", header = F)
colnames(jp) <- c("Y_job_prof_score", "X1_test_score", "X2_test_score", "X3_test_score", "X4_test_score")
glimpse(jp)
```

```
## Rows: 25
## Columns: 5
## $ Y_job_prof_score <dbl> 88, 80, 96, 76, 80, 73, 58, 116, 104, 99, 64, 126, 94~
## $ X1_test_score    <dbl> 86, 62, 110, 101, 100, 78, 120, 105, 112, 120, 87, 13~
## $ X2_test_score    <dbl> 110, 97, 107, 117, 101, 85, 77, 122, 119, 89, 81, 120~
## $ X3_test_score    <dbl> 100, 99, 103, 93, 95, 95, 80, 116, 106, 105, 90, 113,~
## $ X4_test_score    <dbl> 87, 100, 103, 95, 88, 84, 74, 102, 105, 97, 88, 108, ~
```

a. Variable selection using stepwise selection criteria.

```
jP_MLR_allin <- lm(data = jp, Y_job_prof_score ~ .)
step(jP_MLR_allin, direction = "both")

## Start:  AIC=74.95
## Y_job_prof_score ~ X1_test_score + X2_test_score + X3_test_score +
##      X4_test_score
##
##              Df Sum of Sq    RSS    AIC
## - X2_test_score  1      12.22  348.20  73.847
## <none>                        335.98  74.954
## - X4_test_score  1     260.74  596.72  87.314
## - X1_test_score  1     759.83 1095.81 102.509
## - X3_test_score  1    1064.15 1400.13 108.636
##
## Step:  AIC=73.85
## Y_job_prof_score ~ X1_test_score + X3_test_score + X4_test_score
##
##              Df Sum of Sq    RSS    AIC
## <none>                        348.20  73.847
## + X2_test_score  1      12.22  335.98  74.954
## - X4_test_score  1     258.46  606.66  85.727
## - X1_test_score  1     763.12 1111.31 100.861
## - X3_test_score  1    1324.39 1672.59 111.081
##
##
## Call:
## lm(formula = Y_job_prof_score ~ X1_test_score + X3_test_score +
##      X4_test_score, data = jp)
##
## Coefficients:
## (Intercept)  X1_test_score  X3_test_score  X4_test_score
##    -124.2000      0.2963      1.3570      0.5174
```

Findings: The stepwise selection criteria has selected the predictors X1, X3 and X4 to fit the model for job proficiency, based on $AIC = 73.85$, which was the smallest AIC found considering the full model provided initially.

The estimated regression function is:

$$\widehat{job\ proficiency} = -124.2000 + 0.2963X1 + 1.3570X3 + 0.5174X4$$

b. Find/Compute R², adjusted-R², Mallow's Cp, AIC, SBC (i.e., BIC), PRESS for the final model that you selected in part (a) and for the regression model that includes the first-order terms of all 4 predictors.?

I will use the function below provided by Lab 5 practice to compute R², adjusted-R², AIC, BIC and PRESS:

```
selcri<-function(lmout)
{
  n <- length(lmout$fit)
  rsq <- summary(lmout)$r.sq
  adj.rsq <- summary(lmout)$adj.r.sq
  aic <- extractAIC(lmout)[2]
  bic <- extractAIC(lmout, k = log(n))[2]
  press <- sum((lmout$residuals/(1 - hatvalues(lmout)))^2)
  cbind(rsq, adj.rsq, aic, bic, press)
}
```

```
jP_MLR_final <- lm(data = jp, Y_job_prof_score ~ X1_test_score + X3_test_score + X4_test_score)
```

```
paste("Results for the final model found in part (a): ")
```

```
## [1] "Results for the final model found in part (a): "
```

```
selcri(jP_MLR_final)
```

```
##           rsq  adj.rsq      aic      bic  press
## [1,] 0.9615422 0.9560482 73.84732 78.72282 471.452
```

```
paste("Results for the full model with all 4 predictor in it: ")
```

```
## [1] "Results for the full model with all 4 predictor in it: "
```

```
selcri(jP_MLR_allin)
```

```
##           rsq  adj.rsq      aic      bic  press
## [1,] 0.9628918 0.9554702 74.95421 81.04859 518.9885
```

To compute Mallow's Cp:

```
anova(jP_MLR_final)
```

```
## Analysis of Variance Table
##
## Response: Y_job_prof_score
##           Df Sum Sq Mean Sq F value    Pr(>F)
## X1_test_score  1 2395.9   2395.9 144.496 7.054e-11 ***
```

```
## X3_test_score 1 6051.5 6051.5 364.969 9.359e-15 ***
## X4_test_score 1 258.5 258.5 15.588 0.0007354 ***
## Residuals 21 348.2 16.6
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(jP_MLR_allin)
```

```
## Analysis of Variance Table
##
## Response: Y_job_prof_score
##           Df Sum Sq Mean Sq F value    Pr(>F)
## X1_test_score 1 2395.9 2395.9 142.620 1.480e-10 ***
## X2_test_score 1 1807.0 1807.0 107.565 1.708e-09 ***
## X3_test_score 1 4254.5 4254.5 253.259 8.045e-13 ***
## X4_test_score 1 260.7 260.7 15.521 0.00081 ***
## Residuals 20 336.0 16.8
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
n = nrow(jp)

p_final = 4
sse_final = 348.2

p_allin = 5
sse_allin = 336.0
mse_allin = 16.8

mallow_final = sse_final / mse_allin - (n - 2 * p_final)
paste("Mallows' C for the final model found in part (a): ", round(mallow_final, 2))

## [1] "Mallows' C for the final model found in part (a): 3.73"

mallow_allin = sse_allin / mse_allin - (n - 2 * p_allin)
paste("Mallows' C for the full model with all 4 predictor in it: ", mallow_allin)

## [1] "Mallows' C for the full model with all 4 predictor in it: 5"
```

Findings:

- R^2 for both models are similar and considered high, around 96%, meaning that both models explain a large proportion of the variation in job proficiency. However, since the models contain different number of parameters, R^2 is not considered an appropriate measure to compare the models.
- $Adj - R^2$ for both models are also very high and differ by very little from each other. The $Adj - R^2$ for the final model from part (a), with X1, X3 and X4, has a higher $Adj - R^2$ and is preferred.
- Both AIC and BIC for the model from part (a) are smaller than the ones from the full model, once again indicating the model from part (a) is preferred.

- PRESS for the final model from part (a) is smaller and, therefore, preferred as well.
- Mallows' CP for the model in part (a) with 3 predictors is smaller than the one with all 4 predictors in the model and therefore preferred.

c. Use R function `regsubsets()` to conduct the best subset selection. For those model comparison criteria can be found in the output, are their values the same as the ones you calculated in part (c)?.

```
temp <- regsubsets(Y_job_prof_score ~ ., data = jp)
bestsub <- summary(temp)
bestsub
```

```
## Subset selection object
## Call: regsubsets.formula(Y_job_prof_score ~ ., data = jp)
## 4 Variables (and intercept)
##              Forced in Forced out
## X1_test_score FALSE      FALSE
## X2_test_score FALSE      FALSE
## X3_test_score FALSE      FALSE
## X4_test_score FALSE      FALSE
## 1 subsets of each size up to 4
## Selection Algorithm: exhaustive
##              X1_test_score X2_test_score X3_test_score X4_test_score
## 1 ( 1 ) " "              " "              "*"              " "
## 2 ( 1 ) "*"              " "              "*"              " "
## 3 ( 1 ) "*"              " "              "*"              "*"
## 4 ( 1 ) "*"              "*"              "*"              "*"

```

```
data.frame(n.pred = seq(1:4),
           n.par = seq(2:5),
           R2 = bestsub$rsq,
           adj_R2 = bestsub$adjr2,
           SBC = bestsub$bic,
           Mallows_Cp = bestsub$cp
)
```

```
##   n.pred n.par      R2    adj_R2      SBC Mallows_Cp
## 1     1     1 0.8047247 0.7962344 -34.39587  84.246496
## 2     2     2 0.9329956 0.9269043 -57.91831  17.112978
## 3     3     3 0.9615422 0.9560482 -68.57933   3.727399
## 4     4     4 0.9628918 0.9554702 -66.25356   5.000000

```

Findings:

- The `regsubsets()` function can compute: R^2 , $Adj - R^2$, SBC (or BIC) and Mallows' Cp.
- With exception of BSC, all the other values are the same as the values computed above in part (b).